Logan M. Edwards

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EDUCATION

Florida Institute of Technology – Melbourne, FL — Aug 2024 – Present
B.S. in Astronomy & Astrophysics | Second Major: Planetary Science | Minor: Sustainability
GPA: 3.6 | Anticipated Graduation: May 2027
Relevant Coursework: Observational Astronomy, Methods & Instrumentation for Astronomy, Deep Learning (CNNs & PyTorch), Classical Mechanics, Partial Differential Equations, Nanotechnology Lab, Remote Sensing

University of Oxford (Study Abroad) – Oxford, UK — Jun 2025 – Aug 2025
Coursework: Differential Equations, Scientific & Technical Communication

RESEARCH & TECHNICAL PROJECTS

Asteroid Belt Compositional Gradient (Python, AASTeX): Single-author research paper testing the snow-line origin of the S/C-type gradient across 161,659 classified main-belt asteroids; collapsing collisional families and correcting brightness bias moved the crossover from 2.84 to 2.41–2.47 au, favoring implantation from beyond Jupiter.

Stellar Spectral Classification (Python, TensorFlow): Multi-task CNN that predicts Morgan–Keenan spectral classes from Pan-STARRS cutouts, with 84.3% top-1 and 98.4% top-2 accuracy on a 96,039-star multi-survey dataset.

Drone Target Identification (Python, TensorFlow, YOLO, OpenCV): YOLO segmentation front end plus a multi-output CNN classifier, trained on 182,413 images, with 87.1% fine-class accuracy against a 23.5% baseline.

Ortega Telescope Exposure Time Calculator (Python, JavaScript): Photometric calculator that solves exposure times from magnitude, signal-to-noise ratio, seeing, sky brightness, and airmass across five filters.

Asteroid Mining Profitability Pipeline (Python, Streamlit, Docker): Pipeline that catalogs 1.58 million asteroids, prices their minerals, and runs a rocket-equation cost cascade over seven destinations; the most favorable case costs 1.09× its revenue.

Mars Global Mosaic (ArcGIS Pro, ArcPy, GDAL, U-Net): Co-registered Viking, THEMIS, and HRSC/MOLA mosaics over 86.6% of Mars, mapped seven landform classes with SVM and U-Net, and detected 465,354 closed depressions.

EXPERIENCE

AspireCURES – Melbourne, FL
Chief Technology Officer — Aug 2025 – Present
- Direct all web development and technical operations for a private company that connects patients with rare white matter diseases to clinical trials, research studies, and genetic testing.
- Led the migration from Squarespace to Cloudflare Pages; now maintain serverless patient registration, CRM-backed lead handoff, and a twice-weekly research feed across nine disease programs.

Sisters' Hope Foundation – Melbourne, FL
Vice President of Database Management — May 2025 – Present
- Manage the patient database for a 501(c)(3) global rare-disease nonprofit, overseeing its structure and records.
- Migrated the foundation's 167-page website from Squarespace to Cloudflare Pages, with automated link and SEO checks.

Program Support Associate — Jun 2024 – May 2025
- Maintained internal databases and streamlined data workflows, producing reports for program planning and evaluation.
- Led the integration and modernization of the patient database, earning promotion to Vice President.

Intern — Jun 2021 – Jun 2024
- Supported fundraising and outreach, provided general technical support, and assisted patients and their families.

SKILLS

Programming Languages: *Advanced:* Python; *Proficient:* JavaScript, HTML/CSS, LaTeX; *Intermediate:* Bash, PowerShell, Perl, MATLAB

Libraries & Frameworks: PyTorch, TensorFlow/Keras, scikit-learn, NumPy, SciPy, pandas, Astropy, Matplotlib, Streamlit, pytest

Machine Learning: Deep Learning, CNNs, Multi-Task Learning, U-Net Segmentation, YOLO, OpenCV, Computer Vision, SVMs

Astronomy & Data Analysis: Observational Astronomy, Photometry, CCD Instrumentation, Astronomical Catalogs (Gaia DR3, Pan-STARRS1, WISE, JPL SBDB), Catalog Cross-Matching, Bias Correction & Bootstrap Statistics

Orbital Mechanics: SGP4/TLE Propagation, Integrators (Dormand–Prince, RK4, Wisdom–Holman), Perturbations, Delta-v Budgets

GIS & Remote Sensing: ArcGIS Pro, ArcPy, GDAL, Raster Co-Registration, Planetary Image Mosaics, Land-Cover Classification

Linux & Systems: Linux (Ubuntu, Arch), Server Administration, SSH, Shell Scripting, Docker, Git

Web & Data Infrastructure: Cloudflare Pages, Functions & KV, REST APIs, Zoho CRM, SEO, Database Administration & Design

Laboratory Instrumentation: Electron Microscopy (TEM, SEM), Scanning Probe Microscopy (AFM, STM)